Highlights

DaCaF: Bayes-optimal per-metric inference for probabilistic multi-label classification

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- Bayes-optimal per-metric inference for any probabilistic multi-label classifier
- Divide-and-conquer plus fusion: exact optima over two whole metric families
- Seven predict-rules, each verified against brute-force enumeration
- pip-installable and Dockerised, one-command reproduction of paper tables

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ABSTRACT

TODO abstract (<= 250 words).

1. Motivation and significance

- Multi-label classification: each instance carries any subset of L labels.
- Key fact: different evaluation metrics are optimised by different predictions; a model great for F_1 can be poor for subset accuracy.
- Right per-instance choice is the Bayes-optimal prediction (BOP): the $\hat{y}$ maximising the *expected* score of the target metric.
- Computing BOP naively is intractable (2^L candidate predictions).
- Gap: no reusable, exact, multi-metric BOP tool for probabilistic MLC; prior work treats one metric at a time and/or approximates.
- DaCaF closes this: a generic recipe + reference implementation that, given any probabilistic model $P(y \mid x)$, returns the exact BOP for a chosen metric.
- Significance: optimise the metric you actually care about; study metric-mismatch effects without approximation noise.
- Tie to the published paper (Information Fusion 2026, Nguyen et al., 2026): proves correctness for two metric families; this software is the artifact behind it.

2. Software description

2.1. Software architecture

- Installable Python package `dacaf_mlc` (pip / Docker; console script `dacaf-mlc`).
- Core: `ProbabilisticClassifierChainCustom` (PCC) wraps any scikit-learn base estimator; estimates $P(y \mid x)$ via a classifier chain.
- Two algorithmic building blocks behind every inference rule:
  - Divide-and-conquer: partition the 2^L predictions into $L+1$ groups by number of relevant labels; best-in-group by sorting labels by a score; global best = best across groups.
  - Fusion: scores need marginal/pairwise probabilities, estimated by fusing the chain's dependent binary classifiers (ancestral sampling).
- Supporting modules: `evaluation_metrics.py`, `metrics_registry.py`, `pipeline.py` (train / k-fold / run_single), `evaluate.py` (CLI), `arff_dataset.py` (MULAN ARFF + 10-fold CV), `datasets.py`, `chest_xray_dataset/` (NIH feature extractor, [image] extra).
- Vendored `skmultiflow` ClassifierChain base for a self-contained install.

2.2. Software functionalities

- Seven per-metric Bayes-optimal rules: `predict_fmeasure`, `predict_hamming`, `predict_markedness`, `predict_precision`, `predict_npv`, `predict_recall`, `predict_subset`.
- Each returns the prediction maximising the expected value of its target metric; conventions in CONVENTIONS.md.
- Covers two whole metric families at once; flags when a metric's optimum is *trivial* (a metric-choice diagnostic).
- Batched predictor: one `predict_proba` per chain level instead of $N \cdot L \cdot 2^L$; verified numerically equal to brute-force enumeration.
- Reproducible evaluation: CLI + scripts (`reproduce_tabular.sh`, Slurm) over MULAN tabular datasets and an NIH ChestX-ray14 image dataset.
- Correctness harness: every rule checked against brute-force expected-metric enumeration on small cases; CI on Python 3.10/3.12.

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Table 1
Code metadata

Nr.	Code metadata description
C1	Current code version: v1.0.0
C2	Permanent link to code/repository: https://github.com/hxtruong6/inference_probabilistic_mlc Zenodo: TODO software DOI
C3	Permanent link to Reproducible Capsule: TODO Code Ocean capsule
C4	Legal Code License: MIT
C5	Code versioning system used: git
C6	Software code languages, tools, services used: Python (≥3.10); NumPy, SciPy, scikit-learn, pandas, joblib; optional PyTorch / torchvision / torchxrayvision ([image] extra)
C7	Compilation requirements, operating environments, dependencies: `pip install -e .` (core) or `".[image]"`; Linux/macOS; Docker image provided
C8	Link to developer documentation/manual: README.md, CONVENTIONS.md, docs/
C9	Support email for questions: hxtruong@jaist.ac.jp

Table 2
CHD-49, PCC + logistic regression, mean over 5 seeds × 10-fold CV (%, higher is better). Read each column: the bold diagonal entry (optimise the metric you evaluate) is the maximum of its column. NPV and Recall collapse to the predict-all trivial optimum.

Target ↓ / Eval →	F_1	Hamming	Markedness	NPV	Precision	Recall	Subset
Predict F_1	**67.07**	69.29	71.91	81.09	62.74	79.22	15.06
Predict Hamming	63.92	**70.78**	71.27	75.74	66.62	66.93	18.05
Predict Markedness	34.20	63.70	**76.96**	71.10	33.37	40.45	8.39
Predict NPV	58.35	43.01	71.50	**100.00**	43.01	99.46	0.00
Predict Precision	40.54	64.83	68.31	63.10	**73.52**	29.10	3.10
Predict Recall	58.35	43.01	71.50	100.00	43.01	**99.46**	0.00
Predict Subset	63.96	69.37	70.35	76.47	64.15	69.79	**18.92**

3. Illustrative examples

- Minimal library usage (fit once, query any metric's BOP):

```
from sklearn.linear_model import LogisticRegression
from dacaf_mlc.probability_classifier_chains \
    import ProbabilisticClassifierChainCustom
from dacaf_mlc.evaluation_metrics import EvaluationMetrics as EM

pcc = ProbabilisticClassifierChainCustom(
    LogisticRegression(max_iter=10_000))
pcc.fit(X_train, Y_train)            # Y: (n, L) binary
y_f1  = pcc.predict_fmeasure(X_test, beta=1)  # BOP for F1
y_ham = pcc.predict_hamming(X_test)          # BOP for Hamming
print(EM.f_beta(Y_test, y_f1), EM.hamming(Y_test, y_ham))
```

- Headline experiment (mismatch hurts): Table 2, rows = metric optimised for, columns = metric evaluated; bold diagonal = column maximum.
- One command reproduces it: `make reproduce` (CHD-49, fastest dataset).
- The chest-xray path scales the same API to real DenseNet/ResNet features.

4. Impact

- Enables an exact research question: how much does optimising the wrong metric cost, measured without approximation blurring the picture?
- Model-agnostic: works with *any* probabilistic MLC model; a drop-in BOP layer on top of existing $P(y \mid x)$ estimators.
- Breadth: one tool covers two whole metric families; the trivial-optimum diagnostic guides metric selection.
- Reuse beyond the paper: clinical/image MLC (NIH ChestX-ray14 demo), benchmarking, and teaching Bayes-optimal decision-making in MLC.
- Trust: closed-form rules verified against brute-force enumeration; CI keeps it reproducible.
- Low barrier: pip-installable, Dockerised, scripted reproduction, MULAN datasets bundled; a newcomer reproduces a paper table in minutes.
- Extensibility: registry pattern to add metrics/datasets; vendored base keeps installs self-contained.
- Who benefits: MLC researchers, applied ML practitioners choosing a deployment metric, and educators.

5. Conclusions

- DaCaF turns a proved theoretical recipe into a reusable, verified, exact per-metric inference tool for probabilistic MLC.
- Future: more base models, more metric families, optional approximate fast paths.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

TODO statement (or remove if not applicable).

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References

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